Title: AI Solution Implementation at Finance Start-Up, Westlake Financial Ltd.
Word Count: 2,191

1. Business Context

Westlake Financial is a FinTech start-up operating in the banking landscape, offering transactional banking and lending products. The previous report outlined three key areas where AI-augmented solutions can be employed: personalised product recommendations, credit risk assessment, and payment fraud detection (Westlake, 2026). This experiment focuses on credit risk assessment, the process by which a lender estimates the probability of default (PD). Automating this function with a machine learning (ML) model offers significant operational benefits, including reduced decision-making time and lower overhead costs. The models examined include Decision Trees (DTs), Random Forests (RFs), and Adaptive Boosting. Westlake Financial should adopt a risk-averse implementation approach until it accumulates sufficient proprietary data, which will enable the organisation to further fine-tune any model that is initially deployed.

2. Dataset Justification

The Default of Credit Card Clients dataset (Yeh, 2009) is selected for three main reasons. Firstly, it is a real-world banking dataset containing binary labels for payment default (1 = defaulted, 0 = non-defaulted).

Secondly, it contains 23 features (see Appendix 1) that broadly align with the required transactional and customer behavioural information outlined in the previous report (Westlake, 2026, p. 6). However, the following quality issues were identified and addressed during preprocessing. Specifically, EDUCATION variable contains undefined categories (0, 5, 6), MARRIAGE includes an undefined category (0), and the PAY_X variables contain values (-2, 0) not explicitly defined in Yeh's (2009) schema.

Finally, the dataset comprises 30,000 instances with no missing values, enabling statistically robust model evaluation. However, the dataset is imbalanced, with a 78% to 22% distribution between non-defaulted (23,364) and defaulted (6,636) cases (see

Figure 1). While this reflects real-world credit conditions, where default rates in the UK are below 20% for unsecured lending (Bank of England, 2026), the dataset's Taiwanese origin and age (2005) may limit transferability to UK credit conditions.

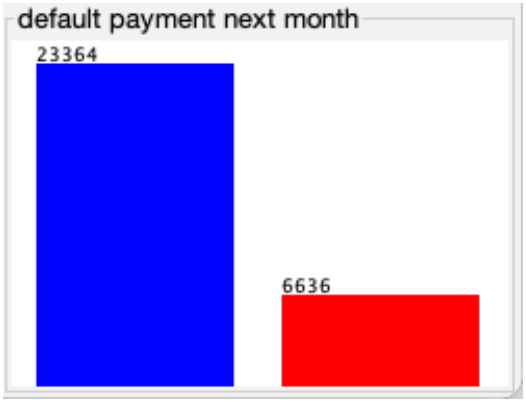


Figure 1: Imbalanced Dataset

3. Modelling Approach

3.1. CRISP-DM Framework

This experiment follows the CRISP-DM framework (see Table 1), which provides a structured and iterative process model suitable for applied ML projects (Schröer, Kruse and Gómez, 2021).

Table 1: CRISP-DM Framework Application

Phase	Experiment Section
Business Understanding	Outlined in Business Context
Data Understanding	Described under Dataset Justification
Data Preparation	Detailed in Modelling Approach under Data Preprocessing
Modelling	Outlined in Modelling Approach and Algorithm Selection
Evaluation	Covered in Evaluation of Experiment Results
Deployment	Detailed in Applicability to Westlake Financial

3.2. Data Preprocessing and Feature Selection

Prior to importing the dataset into WEKA, the PAY_X features were cleansed by converting values of -2 and -1 to 0. According to Yeh (2009), -1 indicates a settled

repayment; therefore, it is assumed that -2 and 0 similarly represent non-overdue repayment behaviour.

Upon import into WEKA, the dataset was converted from CSV to ARFF format. The target variable was transformed to a nominal type using the *NumericToNominal* filter.

The following correlated features were removed prior to training: PAY_4, PAY_5, PAY_6, BILL_AMT4, BILL_AMT5, BILL_AMT6, PAY_AMT4, PAY_AMT5, and PAY_AMT6. Additionally, SEX, EDUCATION, and MARRIAGE were removed on ethical grounds, as credit decisions should not be influenced by protected demographic characteristics.

3.3. Experiment Parameters

3.3.1. Model Overfitting Mitigation

To promote generalisation and reduce overfitting, the following constraints were applied (see Appendices 3-5):

1. A shallow *J48* DT (*confidenceFactor* = 0.1; *minNumObj* = 15)
2. A constrained RF (*minNum* = 5; *maxDepth* = 3; *numIterations* = 100)
3. AdaBoostM1 with *numIterations* = 100 using the shallow *J48* as its base learner

While these constraints improve generalisation, they introduce a risk of underfitting, representing a limitation of the experiment.

3.3.2. Class Imbalance Acceptance (Baseline Learning)

Initial experiments were conducted without addressing class imbalance to preserve the natural distribution of credit data. Overfitting was controlled through model constraints (see 3.3.1), with imbalance effects addressed in Sections 3.3.3 and 3.3.4.

3.3.3. Cost-Sensitive Learning

The experiment was repeated with a cost-sensitive matrix (Table 2) applied across all three classifiers described in Section 3.3.1, reflecting a risk-averse stance in which failing to identify a defaulter is considered five times more costly than a false positive (see Appendix 6). Beyond capturing this asymmetric business cost, the matrix also

serves as an algorithm-level mechanism for managing class imbalance, an approach He and Garcia (2009) recognise as a legitimate alternative to data-level resampling.

Table 2: Cost-Sensitive Matrix

	Predicted 0 (Non-Default)	**Predicted 1 (Default)**
Actual 0 (Non-Default)	0	1
Actual 1 (Default)	5	0

3.3.4. Learning with SMOTE

The experiment was also conducted using the models in Section 3.3.1, applying SMOTE to address class imbalance in the dataset (see Appendix 7). The SMOTE *percentage* was set to 250%, resulting in an approximately 50:50 class distribution with 23,364 non-defaulted and 23,226 defaulted instances (see Figure 2).

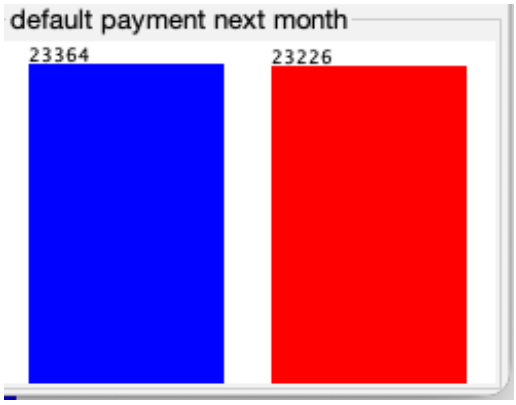


Figure 2: SMOTE-Augmented Dataset

However, a balanced dataset for credit default is unlikely to be representative of real-world conditions. A synthetically balanced dataset may misrepresent real-world default rates, risking miscalibrated PD estimates and reduced regulatory defensibility under the Financial Conduct Authority's (2024) transparency requirements. Consequently, the application of this method should be interpreted with caution, particularly with respect to its implications for the validity and accuracy of the findings.

4. Algorithm Selection

4.1. Decision Tree (DT)

WEKA's *J48* DT, which implements the C4.5 algorithm, was selected for its interpretability and ability to produce transparent, explainable results (Bell, 2020, Chapter 5). The algorithm predicts payment default status, with results readily visualised as a tree diagram. However, DTs are susceptible to overfitting and often result in overly complex models with poor generalisation. DTs are highly unstable as small changes in the training data can result in vastly different tree structures and predictions when applied to the test set. To mitigate these issues, a shallow tree configuration is adopted as the base learner (see Section 3.3.1), constraining model complexity at the cost of increased false positives, as the classifier more aggressively flags potential defaults.

4.2. Random Forest (RF)

RFs are ensemble methods that combine bagging with random feature selection to reduce overfitting and typically yield stronger predictive performance than individual DTs (Kneusel, 2023, Chapter 3). RFs can robustly handle numerical and categorical features while also being less sensitive to noisy attributes, making them well-suited for the dataset's characteristics. However, RFs sacrifice the interpretability of a single DT, producing results that are less readily explainable. This is important in a regulated financial services environment in which model transparency is required. To constrain model complexity and potential overfitting, the parameters outlined in Section 3.3.1 have been applied.

4.3. Adaptive Boosting

AdaBoostM1 was introduced as a supplementary comparator to RF, extending beyond the classifiers proposed in Westlake's (2026) report. Although both are ensemble methods, *AdaBoostM1* employs a boosting strategy in contrast to RF's bagging approach. Rather than training learners independently, boosting builds a strong learner by iteratively training weak base learners on a re-weighted training distribution, yielding improved predictive performance (Géron, 2019, Chapter 7; Schapire, 2013). However, the primary limitation of *AdaBoostM1* is its sensitivity to noisy data. To mitigate this, the shallow J48 configuration described in Section 3.3.1

is used as the base learner, constraining the complexity of each weak learner and limiting its exposure to noise.

5. Evaluation of Experiment Results

5.1. Training Procedures

All models were trained using stratified 10-fold cross-validation, a widely adopted evaluation method for classification (Kohavi, 1995; Hastie, Tibshirani and Friedman, 2009, p. 243). It ensures each observation is used for testing exactly once and for training across the remaining nine folds, enabling robust evaluation across the full dataset. This reduces the risk of evaluation bias emerging from an 80:20 train-test split, lowers variance, and ensures all data points contribute to both training and testing.

5.2. Key Metrics

The key metrics used in this experiment are detailed in Table 3.

Table 3: Key Metrics

Metric	Rationale
Accuracy	Overall correctness of predictions. Useful as a high-level metric but less informative in credit risk when class is imbalanced.
True Positive (TP)	Correctly identified defaulters, enabling risk mitigation (e.g., decline or stricter terms).
False Positive (FP)	Creditworthy applicants incorrectly flagged as risky, leading to lost business.
False Negative (FN)	Missed defaulters. Most costly error, as it results in lending to high-risk borrowers.

True Negative (TN)	Correctly identified low-risk applicants, supporting safe portfolio growth.
Recall on Defaults	Measures how well the model identifies actual defaulters. Critical for minimising credit losses.
Precision on Non-Defaults	The proportion of applicants predicted as non-default by the model that genuinely pose low credit risk.
F1-Score	The harmonic mean of Precision and Recall.
AUC-ROC	Measures the model's ability to discriminate between defaulters and non-defaulters across all classification thresholds, where 1.0 indicates perfect separation and 0.5 indicates no better than chance.

5.3. Evaluation of Results

5.3.1. Baseline Learning Models

Although high Precision values suggest surface-level accuracy, all three models exhibit a pronounced bias towards the majority non-default class (see Section 2). Default Recall ranges from 0.318 to 0.371, meaning each model fails to identify between 63% and 68% of actual defaulters, which is a critical shortcoming, given that minimising missed defaults is the primary objective of any credit risk classifier.

Table 4: Baseline Learning Models (see Appendices 3-5)

Metric	Class	J48	RF	AdaBoostM1
TP Rate	Non-Default	0.962	0.963	0.922
	Default	0.320	0.318	0.371
	Weighted Avg.	0.820	0.820	0.800
FP Rate	Non-Default	0.680	0.682	0.629
	Default	0.038	0.037	0.078
	Weighted Avg.	0.538	0.540	0.507

Recall	Non-Default	0.962	0.963	0.922
	Default	0.320	0.318	0.371
	Weighted Avg.	0.820	0.820	0.800
Precision	Non-Default	0.833	0.832	0.878
	Default	0.704	0.707	0.575
	Weighted Avg.	0.804	0.805	0.780
F1-Score	Non-Default	0.893	0.893	0.878
	Default	0.440	0.438	0.451
	Weighted Avg.	0.793	0.792	0.783
AUC-ROC	Non-Default	0.638	0.765	0.739
	Default	0.638	0.765	0.739
	Weighted Avg.	0.638	0.765	0.739

AUC-ROC scores indicate that RF and *AdaBoostM1* achieve fair discriminative performance between classes, while *J48* performs poorly in comparison, sitting in the 0.60-0.70 range.

5.3.2. Cost-Sensitive Learning Models

The introduction of the 5:1 cost matrix produces the intended effect on classifier behaviour (see Table 5). Default Recall increases across all three models with *J48* at 0.750, RF at 0.715, and *AdaBoostM1* at 0.670, reflecting the penalty structure imposed on the default class. This is a key trade-off of credit risk models where an increase in false positives among predicted non-defaulters is accepted in exchange for improved detection of genuine defaults. However, *AdaBoostM1* achieved the lowest Default Recall gain (0.371 to 0.670) and the weakest AUC-ROC of the three models (0.735), confirming that it offers no material advantage over the primary candidates.

Table 5: Cost-Sensitive Learning Models (see Appendix 6)

Metric	Class	J48	RF	AdaBoostM1
TP Rate	Non-Default	0.610	0.670	0.676
	Default	0.750	0.715	0.670
	Weighted	0.641	0.680	0.675
FP Rate	Non-Default	0.250	0.285	0.330
	Default	0.390	0.330	0.324

	Weighted	0.281	0.295	0.328
Recall	Non-Default	0.610	0.670	0.676
	Default	0.750	0.715	0.670
	Weighted	0.641	0.680	0.675
Precision	Non-Default	0.896	0.892	0.878
	Default	0.353	0.381	0.370
	Weighted	0.766	0.779	0.766
F1-Score	Non-Default	0.726	0.765	0.764
	Default	0.480	0.497	0.477
	Weighted	0.671	0.706	0.700
AUC-ROC	Non-Default	0.752	0.763	0.735
	Default	0.752	0.763	0.735
	Weighted	0.752	0.763	0.735

Non-Default Precision remains relatively stable compared to the Baseline Learning Models; however, Default Precision declines significantly, indicating that creditworthy applicants are incorrectly flagged as potential defaulters. This is the expected result of imposing cost-sensitive learning rather than indicating model degradation.

5.3.3. Learning Models with SMOTE

This method achieves the strongest balance across evaluation metrics. Default Recall ranges from 0.698 to 0.816, which is similar to cost-sensitive learning. Notably, Default Precision is substantially higher under SMOTE (0.818–0.857) than under cost-sensitive learning (0.353–0.381), suggesting that the resampling approach substantially reduces false positive classifications. This improvement is further reflected in the Default F1 scores, which range from 0.753 to 0.836 under SMOTE compared to 0.477–0.497 under cost-sensitive learning. AUC-ROC peaks at 0.911 for *AdaBoostM1* with SMOTE, the highest value recorded across all experimental conditions.

Table 6: Learning Models with SMOTE (see Appendix 7)

Metric	Class	J48	RF	AdaBoostM1
TP Rate	Non-Default	0.882	0.845	0.864
	Default	0.715	0.698	0.816
	Weighted Avg.	0.799	0.772	0.840
FP Rate	Non-Default	0.285	0.302	0.184
	Default	0.118	0.155	0.136
	Weighted Avg.	0.201	0.228	0.160
Recall	Non-Default	0.882	0.845	0.844
	Default	0.715	0.698	0.816
	Weighted Avg.	0.799	0.772	0.840
Precision	Non-Default	0.757	0.738	0.826
	Default	0.858	0.818	0.857
	Weighted Avg.	0.807	0.778	0.841
F1-Score	Non-Default	0.815	0.788	0.844
	Default	0.780	0.753	0.836
	Weighted Avg.	0.798	0.771	0.844
AUC-ROC	Non-Default	0.852	0.837	0.911
	Default	0.852	0.837	0.911
	Weighted Avg.	0.852	0.837	0.911

5.4 Credit Risk Narrative

Table 7 indicates that the imbalanced dataset models outperform the alternative models due to their higher overall accuracy.

Table 7: Model Accuracy (See Appendices 3-7)

Metric	J48	RF	AdaBoostM1
Imbalanced Dataset	81.9867%	81.9933%	80.0233%
Cost-Sensitive	64.0767%	68.0133%	67.4533%

SMOTE	79.9056%	77.199%	84.0331%

However, Section 5.3 demonstrates these models are the least robust in terms of overall predictive performance as accuracy alone fails to capture the distributional challenges inherent in imbalanced credit datasets.

SMOTE yields the strongest overall performance; however, these results were evaluated on a rebalanced dataset where oversampling was applied prior to cross-validation, introducing a risk of data leakage. The observed performance differential between SMOTE and cost-sensitive learning (e.g., RF AUC-ROC of 0.837 in Table 6 versus 0.763 in Table 5) should not be interpreted as a genuine performance gap but rather as the result of the inflated training distribution. Therefore, the SMOTE results represent an upper-bound performance estimate given that oversampling preceded cross-validation partitioning. These results should be treated as an optimistic performance boundary rather than a deployable benchmark. Appendix 10 demonstrates another method for applying SMOTE to *J48*, which generated less optimistic results.

The cost-sensitive learning models show the most operationally realistic results as they address the asymmetric misclassification costs inherent in imbalanced credit datasets. However, their primary limitation is an elevated false positive rate where creditworthy applicants are predicted as defaulters, which may consequently be denied credit. This underscores the trade-offs discussed in Section 5.3.2, whereby the cost of an undetected default is substantially greater than the cost of declining a creditworthy customer.

6. Applicability to Westlake Financial

6.1. Initial Deployment Phasing

Westlake's (2026) report identified DTs and RFs as candidate algorithms for credit risk assessment. This experiment validated both learning systems under controlled conditions using a real-world credit dataset, with *AdaBoostM1* included as a supplementary ensemble comparator to further stress-test the results. *AdaBoostM1*'s

internal re-weighting mechanism means it exhibits some native sensitivity to minority-class instances, arguably diminishing the incremental benefit of an external cost matrix; in practice, the results in Section 5 confirm that it offers no material advantage over DTs and RFs. For a regulated FinTech start-up such as Westlake Financial, it is recommended that a cost-sensitive DT is initially deployed to augment human decision-making. Its inherent interpretability supports compliance with the FCA's transparency and explainability principle (2024, pp. 15-16), and enables clear justification of credit decisions for internal stakeholders.

6.2. Future Deployment and Conclusion

As Westlake Financial matures and accumulates proprietary customer data, a hybrid approach is recommended: RF as the primary model, augmented by a DT explainability layer. In all configurations, model output should support, rather than replace, human credit decisions. This is supported by the results detailed in Section 5.3.2, where both models demonstrate comparable performance under cost-sensitive conditions. Both models should be retrained on proprietary data to further improve predictive performance. The 5:1 cost-sensitive ratio reflects a conservative starting position appropriate for an early-stage lender; however, as default data accumulates, the cost ratio (see Appendix 9) should be reviewed and empirically derived from Westlake Financial's own loss experience, enabling the models to be more precisely calibrated to the organisation's evolving risk appetite. Experimentation with deeper DTs and tuned RFs (see Appendix 8) is suggested to improve upon the metrics reported in Section 5.3, which were generated using a shallow DT. Finally, post-hoc interpretability techniques, including SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), should be considered as these would provide feature-level attribution for individual credit decisions.

Furthermore, the application of unsupervised methods such as Expectation-Maximisation (EM) clustering could be explored as a complementary extension in future work, enabling Westlake Financial to derive latent customer profiles that supplement the aforementioned supervised credit risk models. EM clustering could further streamline credit decisions by routing high-risk and low-risk customers to automated decisions and ambiguous profiles to manual review.

In conclusion, the experiment's findings support a controlled, governance-led adoption of AI in credit risk assessment, where interpretability, regulatory compliance, and operational scalability are prioritised alongside predictive performance. A layered modelling architecture that combines DTs for transparency and RFs for robust scoring provides the most effective and scalable pathway for Westlake Financial to integrate AI into its lending operations.

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Appendices

Appendix 1 — Explanation of the features in Yeh's (2009) Dataset

- Table 8 details the 23 features in Yeh's (2009) dataset, describing each feature to contextualise the removal of correlated and ethically sensitive attributes outlined in Section 3.2.
- The dataset is available from: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

Table 8: Dataset Features (Yeh, 2009)

Variable	Description	Explanation
X1	LIMIT_BAL	The amount of credit granted
X2	SEX	Gender (1 = male; 2 = female)
X3	EDUCATION	Education (1 = graduate school; 2 = university; 3 = high school; 4 = others)
X4	MARRIAGE	Marital status (1 = married; 2 = single; 3 = others)
X5	AGE	Age (years)
X6	PAY_0	History of past repayment from most recent to oldest
X7	PAY_2	
X8	PAY_3	
X9	PAY_4	
X10	PAY_5	
X11	PAY_6	
X12	BILL_AMT1	Amount of bill settled from most recent to oldest
X13	BILL_AMT2	
X14	BILL_AMT3	
X15	BILL_AMT4	
X16	BILL_AMT5	
X17	BILL_AMT6	
X18	PAY_AMT1	Amount paid from most recent to oldest bill.
X19	PAY_AMT2	
X20	PAY_AMT3	
X21	PAY_AMT4	
X22	PAY_AMT5	
X23	PAY_AMT6	

Appendix 2 — Feature Selection and Data Cleansing in WEKA

- Figure 3 illustrates the 12 features retained following the removal of correlated repayment history variables (PAY_4–PAY_6), correlated billing and payment amount variables (BILL_AMT4–6, PAY_AMT4–6), and the protected demographic characteristics SEX, EDUCATION, and MARRIAGE:

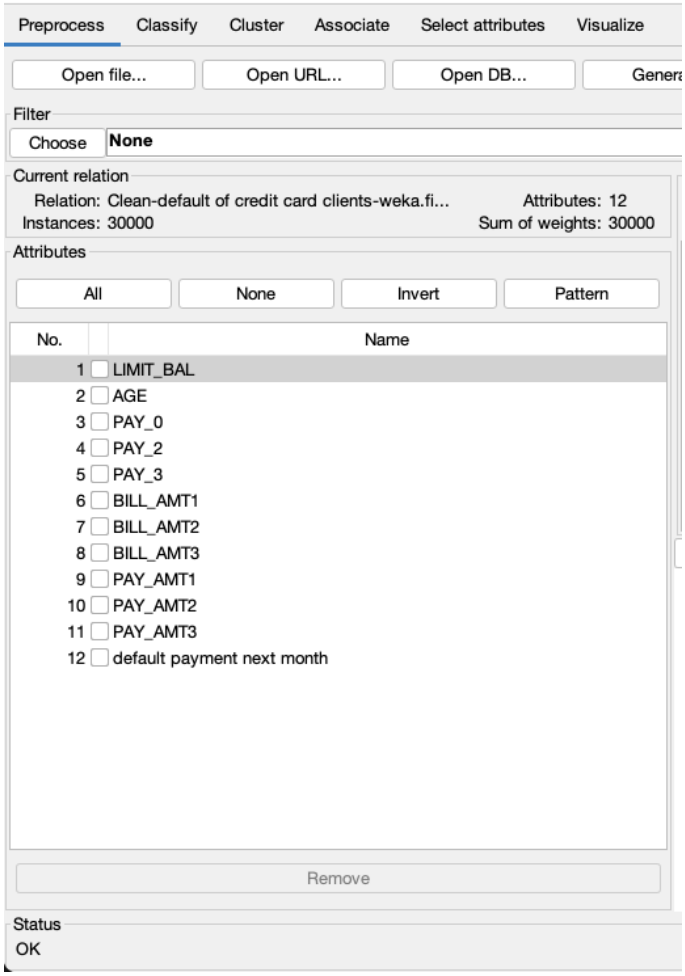


Figure 3: Feature Selection

- Figure 4 illustrates the application of *NumericToNominal* filter, which is used to convert the default and non-default classes to a nominal value, which is required for the tree-based ensemble learning methods reviewed in this experiment.

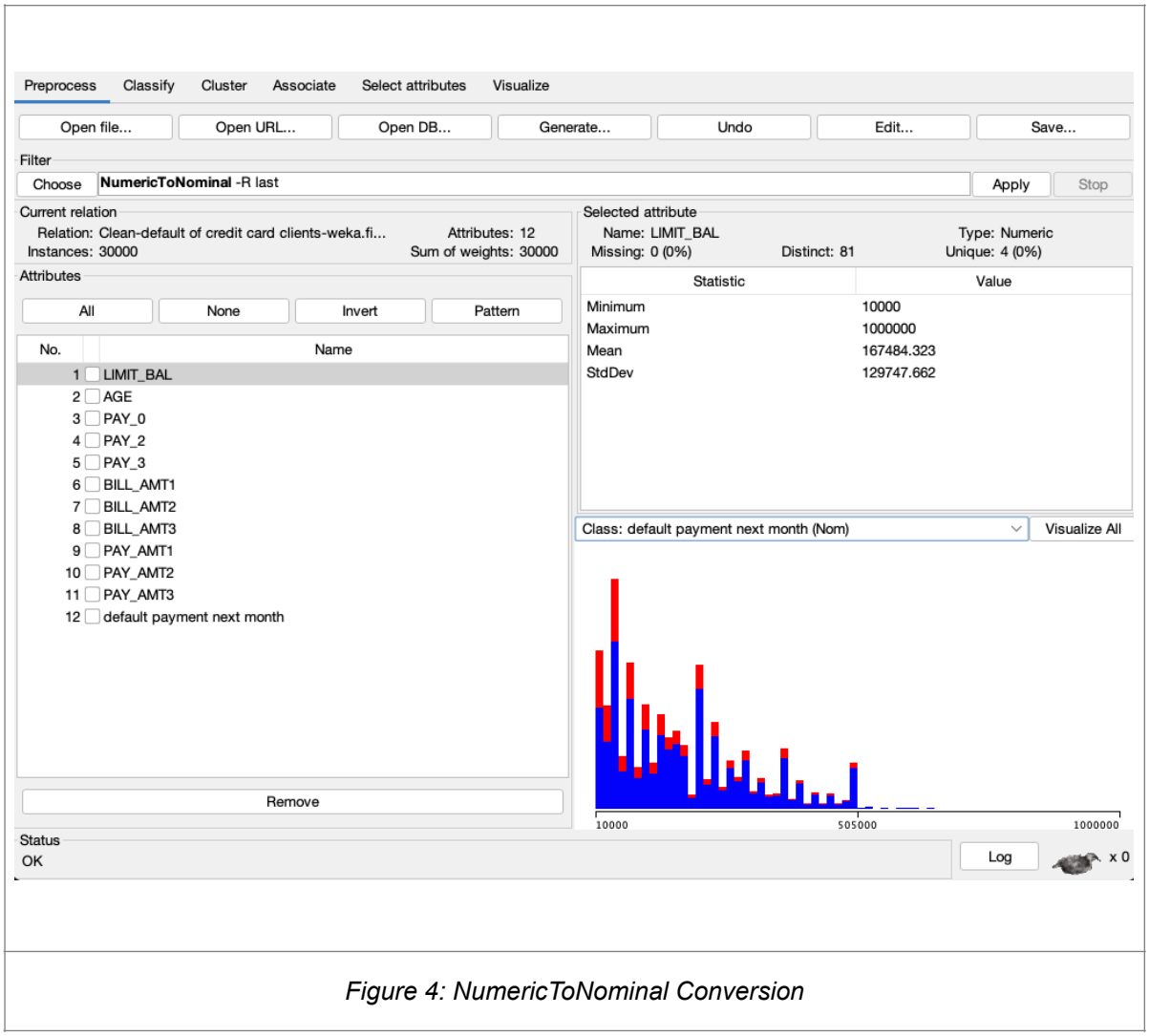


Figure 4: NumericToNominal Conversion

Appendix 3 — Shallow *J48* Decision Tree

- Figure 5 shows the *J48* configuration, with *confidenceFactor* reduced to 0.1 and *minNumObj* increased to 15. A lower *confidenceFactor* increases pruning that reduces tree complexity and overfitting risks; the *minNumObj* configuration prevents splitting on insufficiently populated tree nodes.



Figure 5: J48 Configuration

- Figure 6 presents the classifier output for the baseline *J48* experiment:

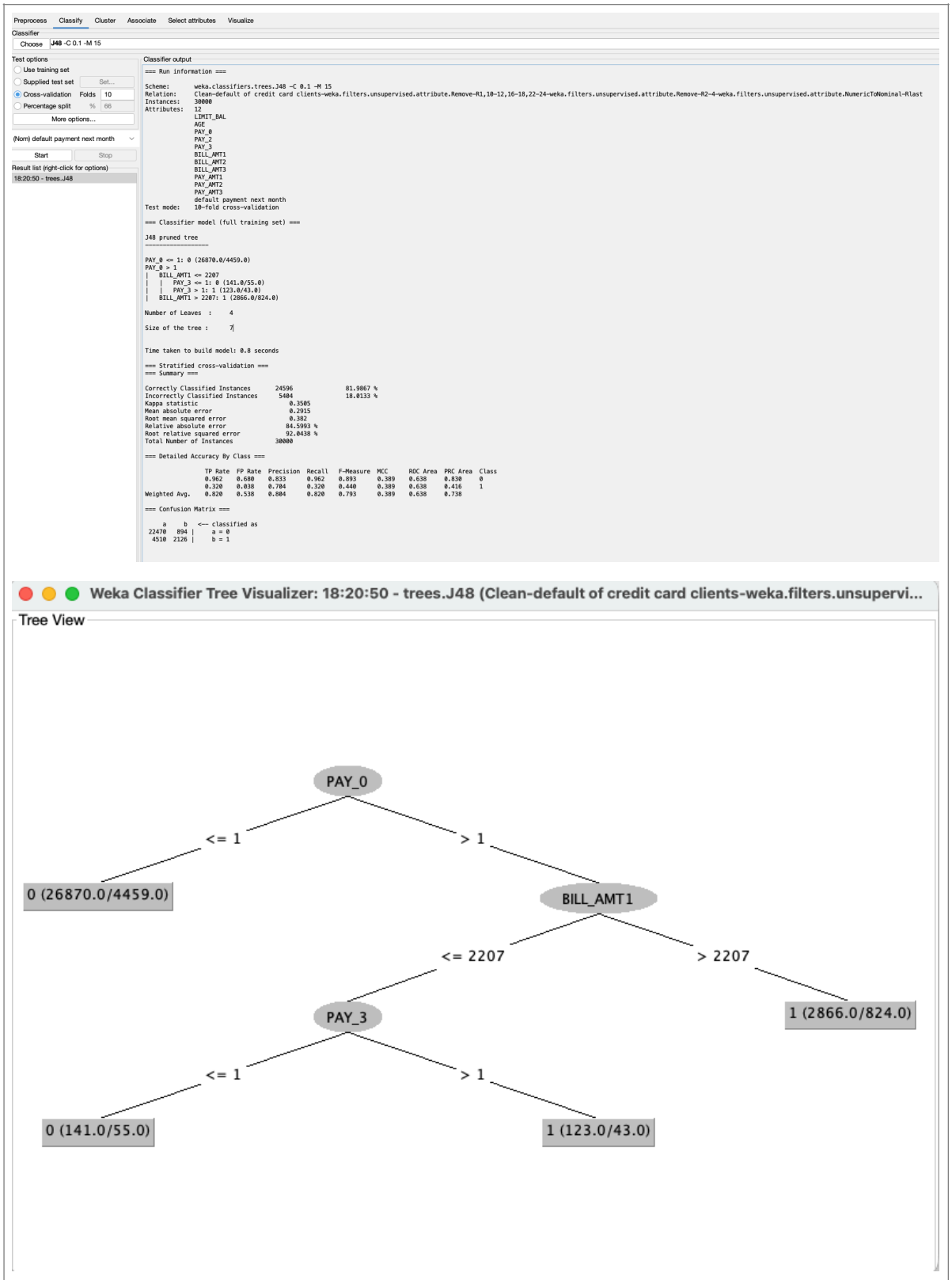


Figure 6: J48 Outputs

Appendix 4 — Random Forest Model

- Figure 7 shows the Random Forest configuration, constrained to a maximum depth of 3 and a minimum of 5 instances per leaf, with 100 iterations. The depth constraint limits each individual tree's complexity, preserving the variance-reduction benefit of bagging while preventing any single learner from memorising majority-class patterns.

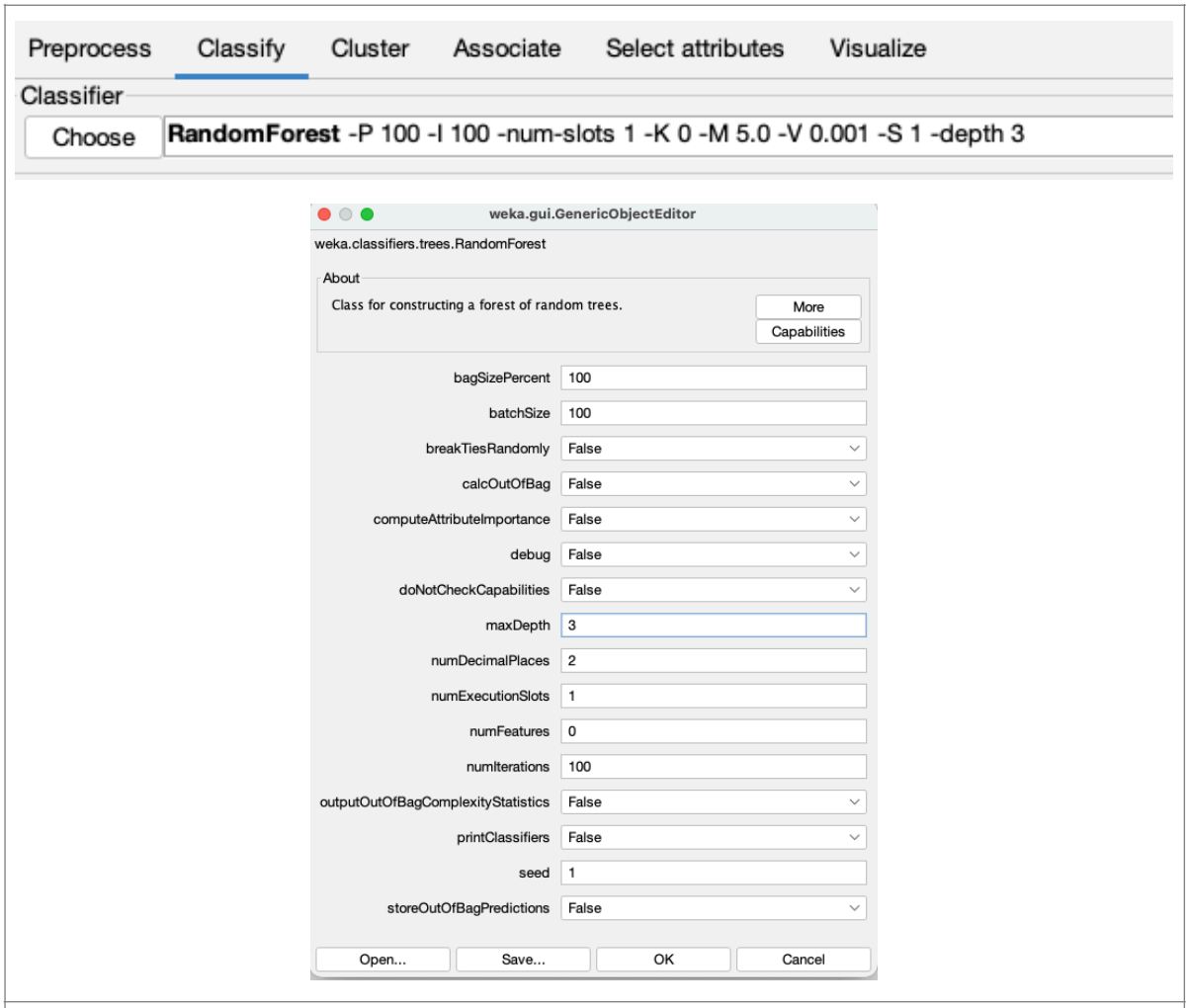


Figure 7: Random Forest Configuration

- Figure 8 presents the baseline Random Forest classifier output.

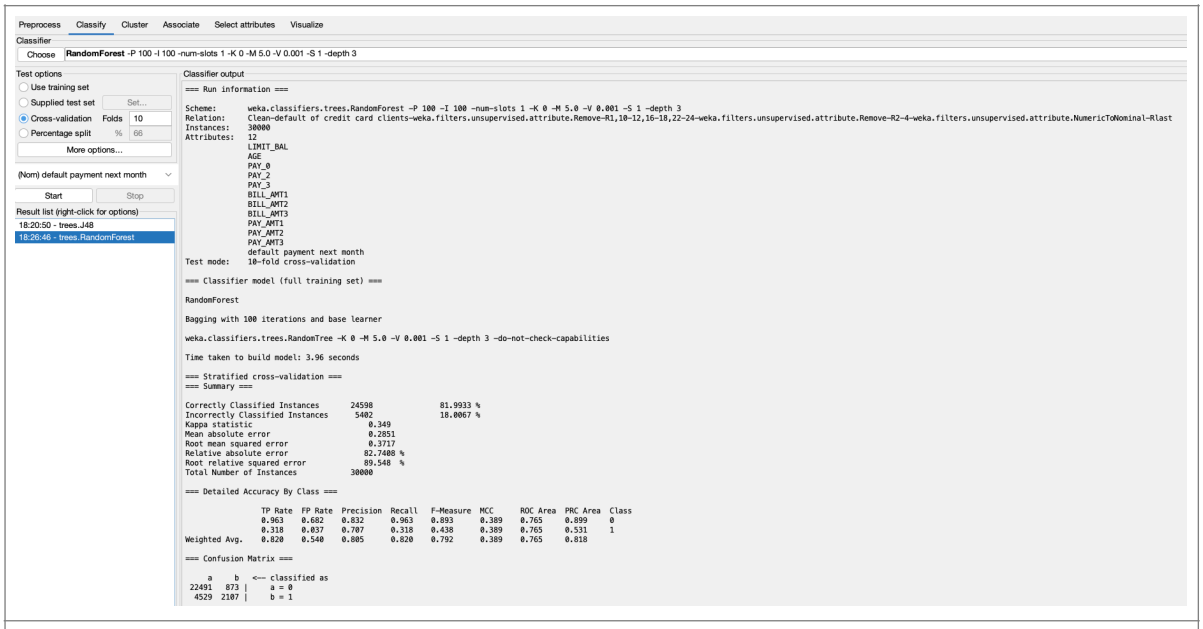


Figure 8: Random Forest Outputs

Appendix 5 — *AdaBoostM1* Model

- Figure 9 shows the *AdaBoostM1* configuration, parameterised with 100 iterations and the shallow *J48* described in Appendix 3 as its base learner. The shallow *J48* is selected deliberately: boosting iteratively reweights misclassified instances, and a complex base learner would absorb noise in early iterations, destabilising the ensemble.

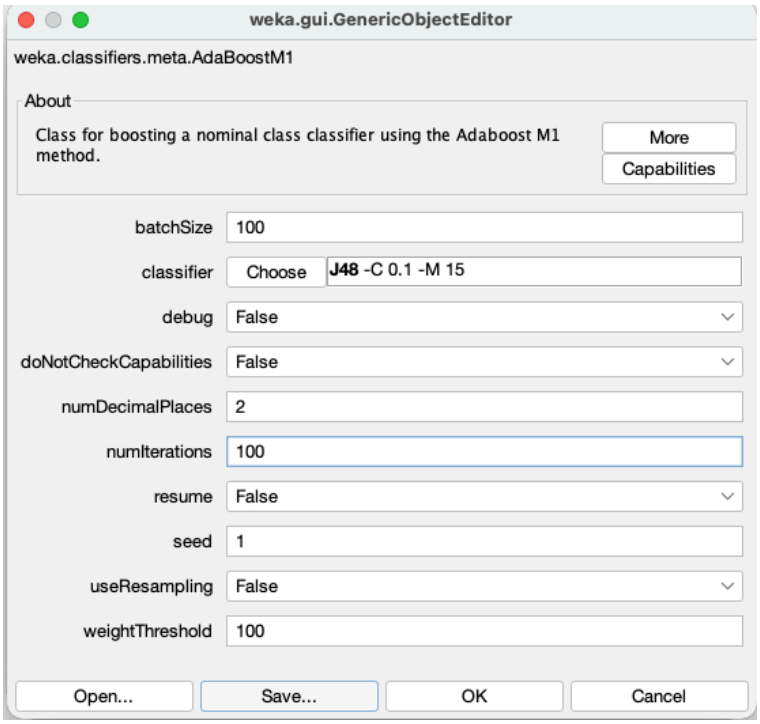


Figure 9: AdaBoostM1 Configuration

- Figure 10 presents the baseline *AdaBoostM1* output. A key observation is that the total *numIterations* capped out at 15, indicating that the model may have correctly classified all training instances for the *J48* base learner which is a relatively weak learner (i.e., a shallow DT).

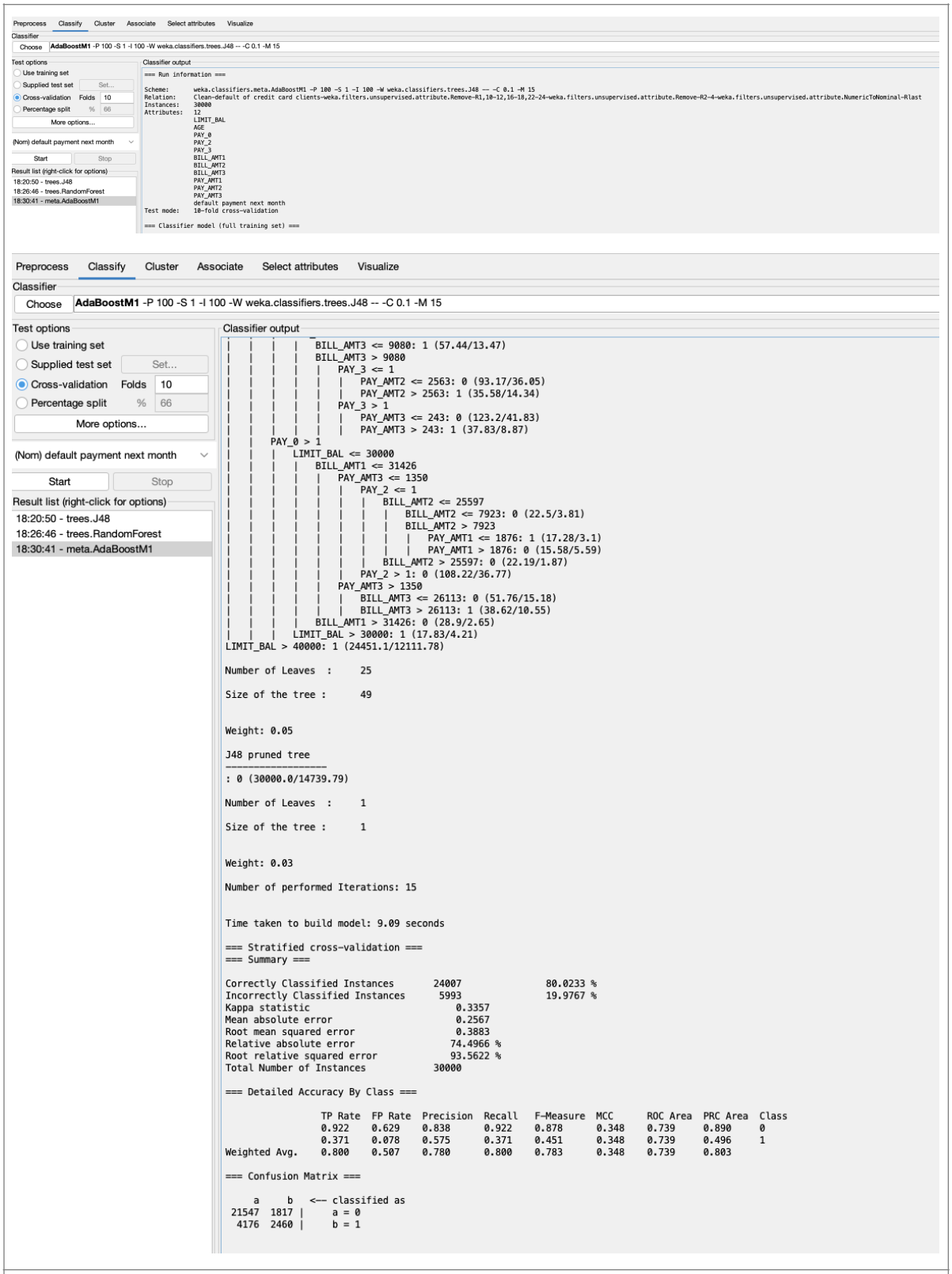


Figure 10: AdaBoostM1 Outputs

Appendix 6 — *Cost-Sensitive Learning*

- The following outputs apply the 5:1 cost-sensitive matrix detailed in Table 2 across all three classifiers.
- *The J48* outputs with a 5:1 cost-sensitive ratio are shown in Figure 11. When comparing this to Figure 6 in Appendix 3, it is clear that the DT generated under these test conditions is more complex:



Figure 11: J48 with Cost-Sensitive Learning

- Random Forest outputs with a 5:1 cost-sensitive ratio are shown in Figure 12:

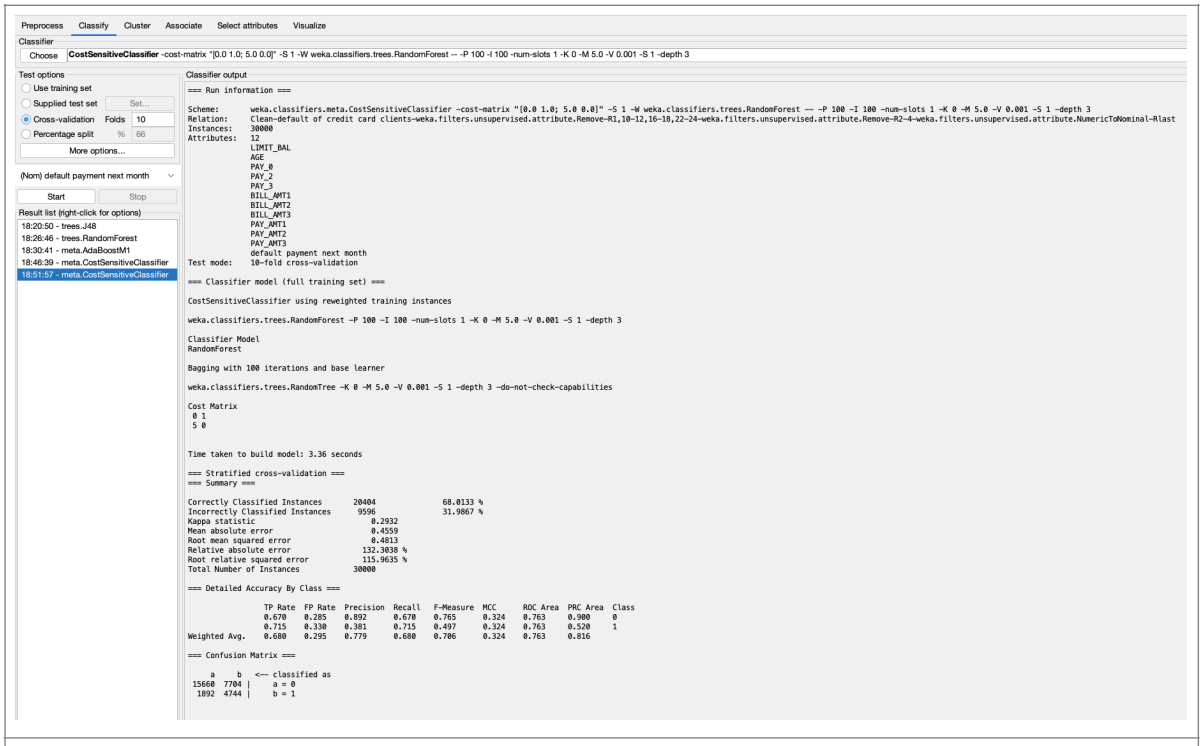


Figure 12: Random Forest with Cost-Sensitive Learning

- *AdaBoostM1* outputs with a 5:1 cost-sensitive ratio are shown in Figure 13:

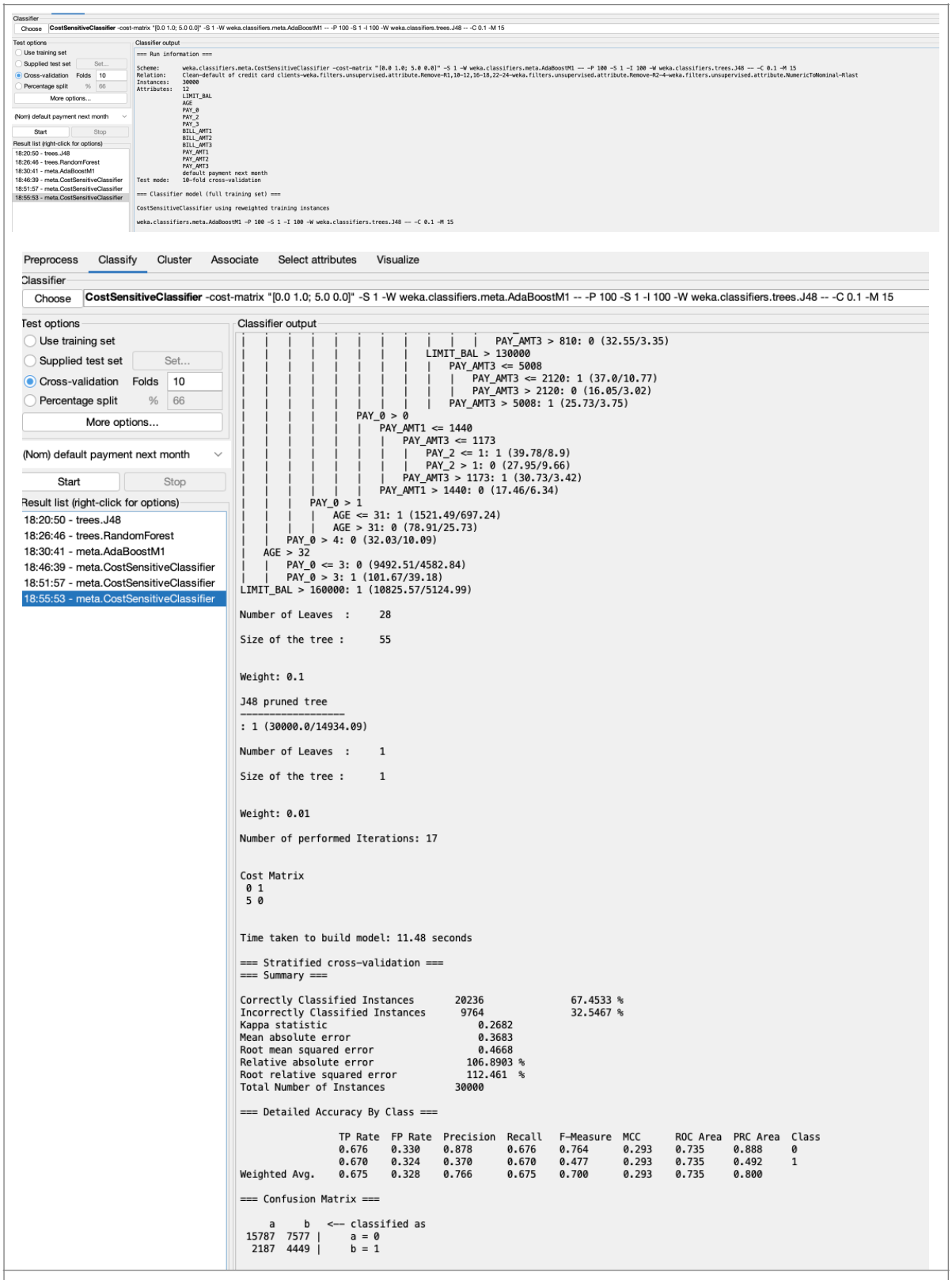


Figure 13: AdaBoostM1 with Cost-Sensitive Learning

Appendix 7 — Application of SMOTE

- SMOTE was applied at 250% to generate a relatively balanced class distribution prior to model training. *Randomise* filter was applied subsequently to prevent any ordering bias introduced by the sequential oversampling process.

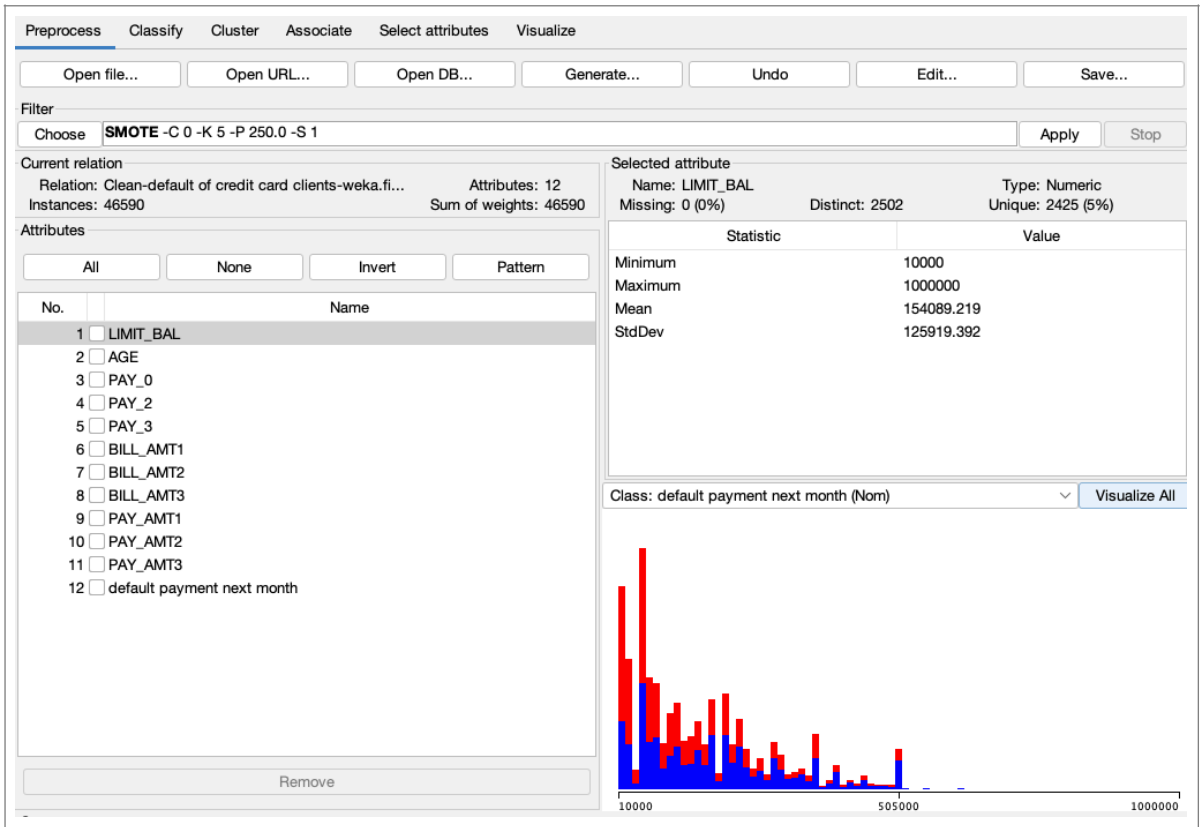


Figure 14: SMOTE

- *J48* with SMOTE and Randomise outputs are shown in Figure 15:

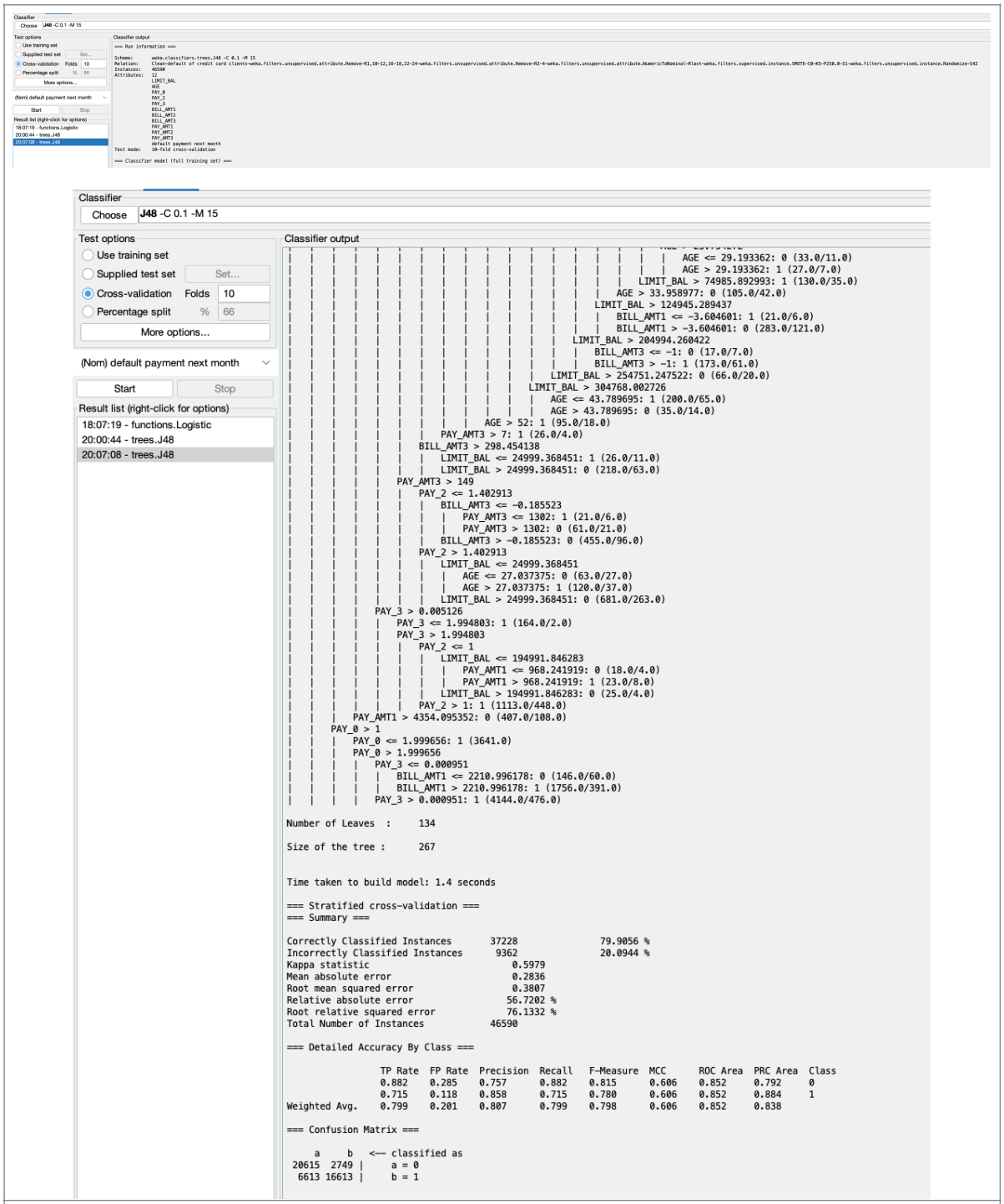


Figure 15: J48 with SMOTE

- Random Forest with SMOTE and Randomise outputs are shown in Figure 16:



Figure 16: Random Forest with SMOTE

- *AdaBoostM1* outputs with SMOTE and Randomise are shown in Figure 17:



Figure 17: AdaBoostM1 with SMOTE

Appendix 8 — Supplementary Testing — Deeper Tree and Random Forest

- Figure 18 presents the deeper *J48* configuration and output, with *minNumObj* reduced to 2 and the cost-sensitive 5:1 matrix retained:

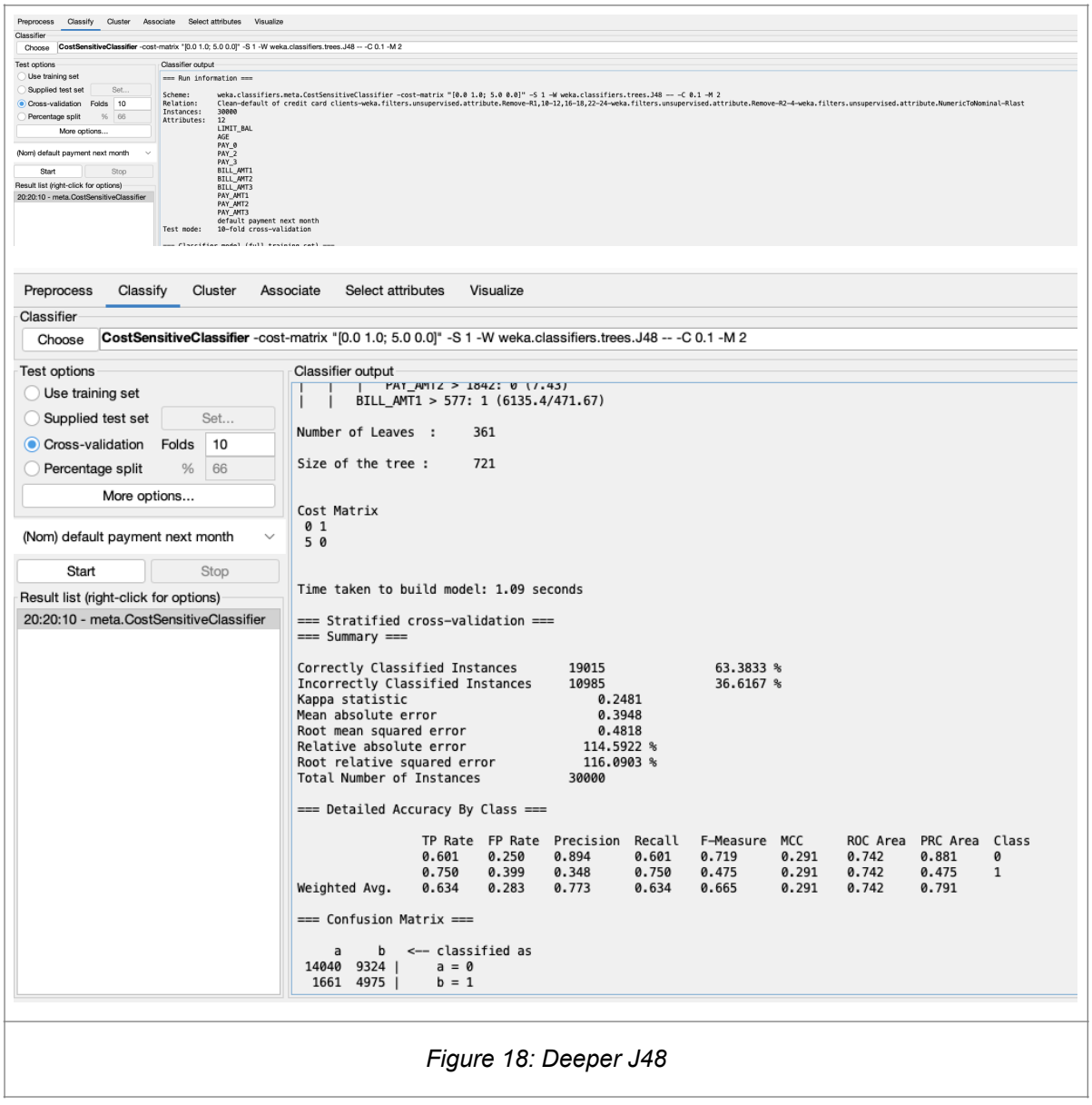


Figure 18: Deeper J48

- Figure 19 presents the deeper Random Forest output with depth constraints removed:

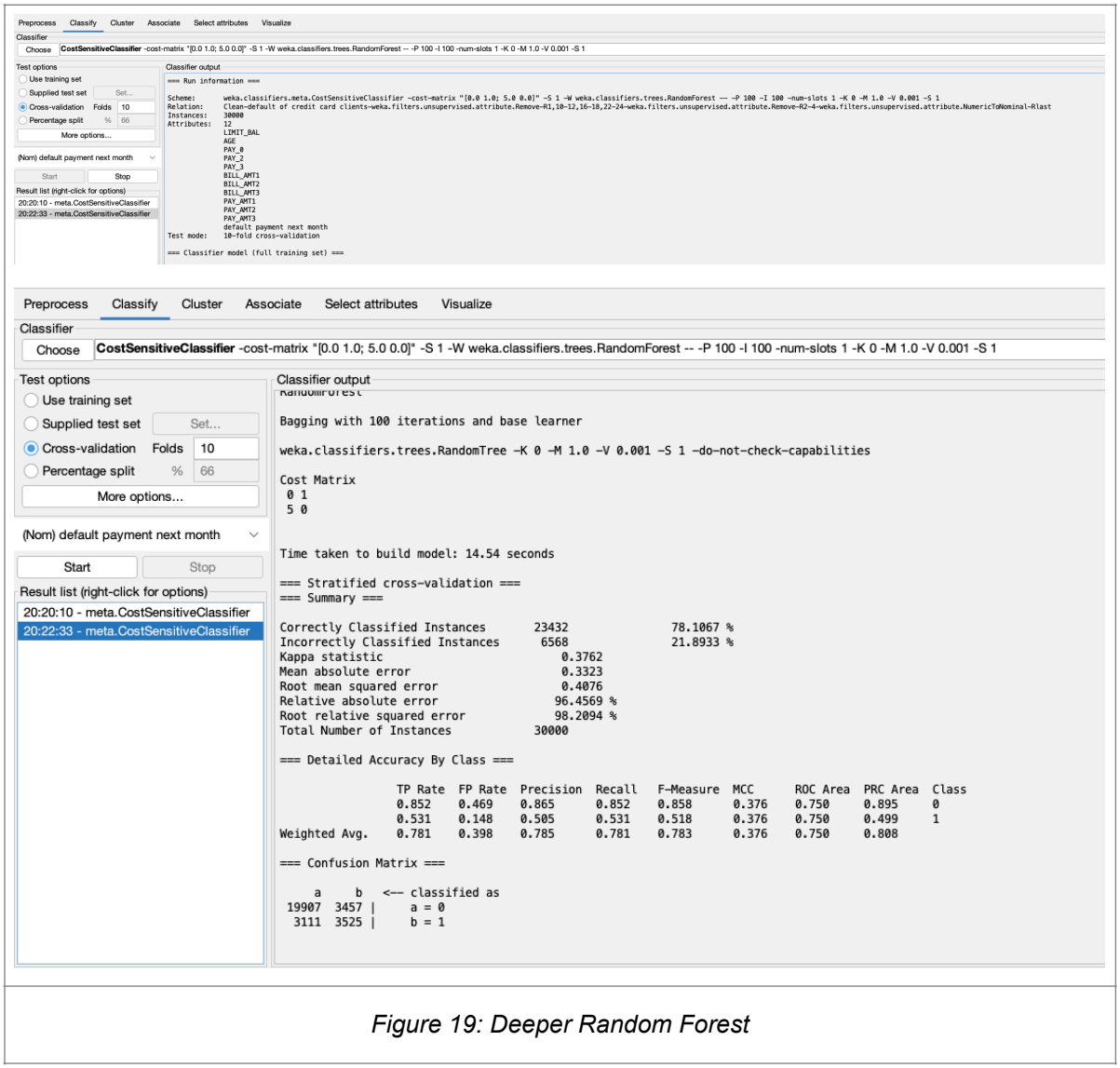


Figure 19: Deeper Random Forest

- Table 9 indicates that a change in the test parameters results in very different behaviour when compared to the results from Section 5.3.2. For example, RF Default Recall is lower at 0.531 (see Table 9) compared to 0.715 (see Table 5) in the original test. This suggests that the *maxDepth* configuration used in this experiment is integral for the model's sensitivity towards defaulters.

Table 9: Key Metrics of Deeper J48 and RF

Metric	Class	J48	RF
TP Rate	Non-Default	0.601	0.852
	Default	0.750	0.531
	Weighted Avg.	0.634	0.781
FP Rate	Non-Default	0.250	0.469
	Default	0.399	0.148
	Weighted Avg.	0.283	0.393
Recall	Non-Default	0.601	0.852
	Default	0.750	0.531
	Weighted Avg.	0.634	0.781
Precision	Non-Default	0.894	0.865
	Default	0.348	0.505
	Weighted Avg.	0.733	0.785
F1-Score	Non-Default	0.719	0.868
	Default	0.475	0.519
	Weighted Avg.	0.665	0.783
AUC-ROC	Non-Default	0.742	0.750
	Default	0.742	0.750
	Weighted Avg.	0.742	0.750

Appendix 9 — Supplementary Testing — Adjusted Cost-Sensitive Ratio

- This approach adjusts the cost-sensitive ratio to 3:1, showcasing that these changes result in differing model behaviour when compared with Table 5 in Section 5.3.2. Figures 20-22 show the configuration and output while Table 10 presents the metrics. Default Recall performs poorly in contrast to Table 5 with values ranging from 0.512 to 0.583, showcasing that changes to the cost-sensitive matrix will result in the model being less able to predict defaults.

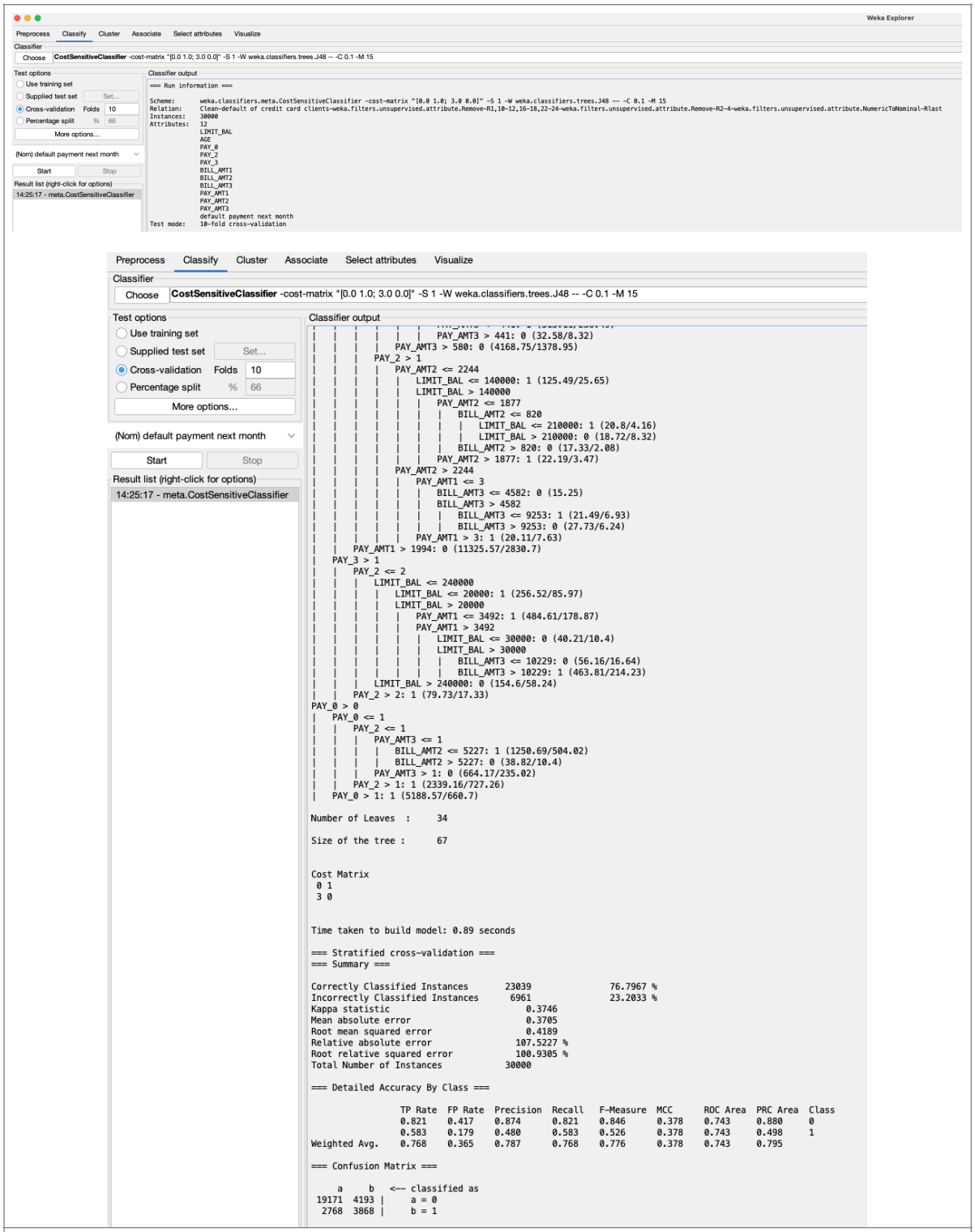


Figure 20: J48 with 3:1 Cost Sensitive Matrix

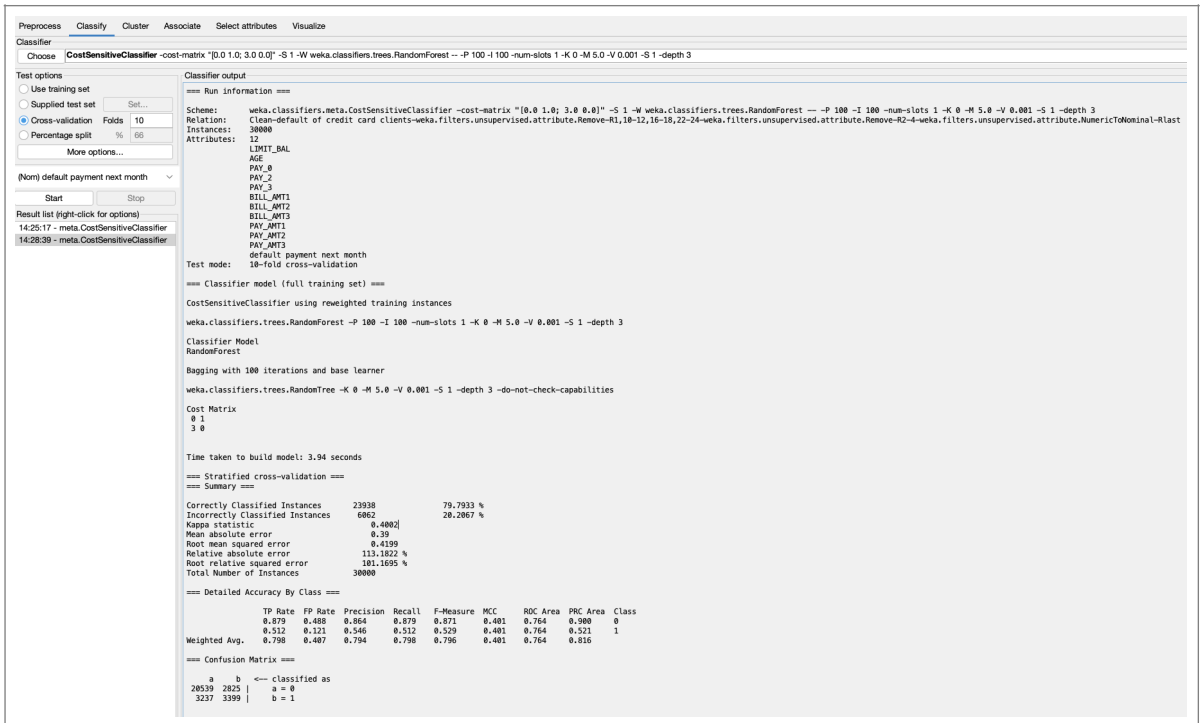


Figure 21: Random Forest with 3:1 Cost Sensitive Matrix

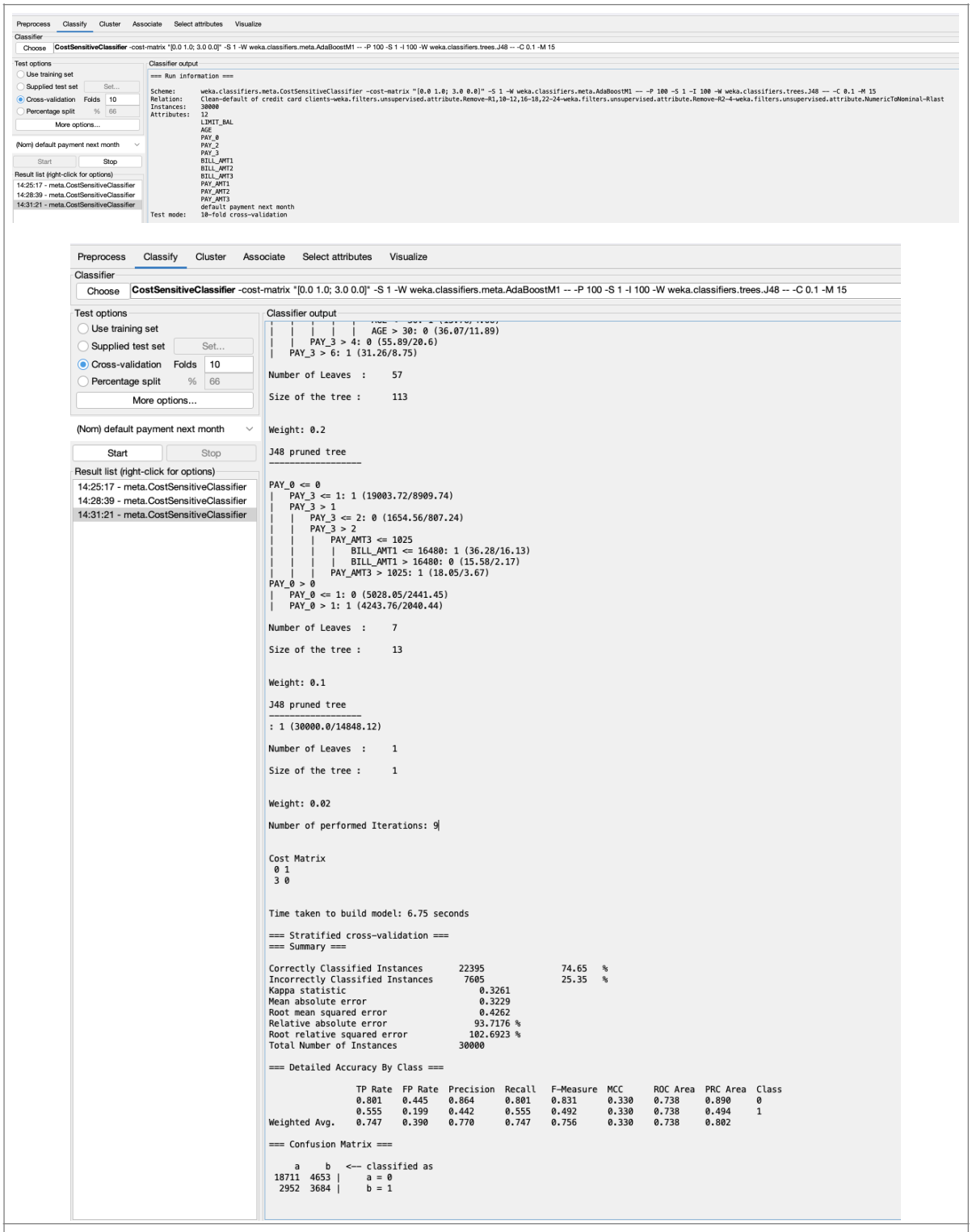


Figure 22: AdaBoostM1 with 3:1 Cost Sensitive Matrix

Table 10: Cost-Sensitive Learning Models with a 3:1 ratio

Metric	Class	J48	RF	AdaBoostM1
TP Rate	Non-Default	0.821	0.879	0.801
	Default	0.583	0.512	0.555
	Weighted Avg.	0.768	0.798	0.747
FP Rate	Non-Default	0.179	0.488	0.445
	Default	0.179	0.121	0.199
	Weighted Avg.	0.365	0.407	0.390
Recall	Non-Default	0.821	0.879	0.801
	Default	0.583	0.512	0.555
	Weighted Avg.	0.768	0.798	0.747
Precision	Non-Default	0.874	0.864	0.864
	Default	0.480	0.546	0.442
	Weighted Avg.	0.787	0.794	0.770
F1-Score	Non-Default	0.846	0.871	0.831
	Default	0.526	0.512	0.492
	Weighted Avg.	0.776	0.796	0.756
AUC-ROC	Non-Default	0.743	0.764	0.738
	Default	0.743	0.764	0.738
	Weighted Avg.	0.743	0.764	0.738

Appendix 10 — Supplementary Testing — Apply SMOTE and classifier application inside *FilteredClassifier*

- This approach wraps SMOTE and the default classifier in *FilteredClassifier*, which applies the filter within each fold during cross-validation. This method prevents synthetic data leakage (see Section 5.4). An additional test has been conducted on the shallow *J48* that has been used throughout this study. Figure 23 below shows the configuration and the outputs, while Table 11 shows the results, which differ significantly from Table 6 in Section 5.3.3. For example, Default Recall is lower at 0.310 when compared to the *J48 SMOTE* value of 0.715 in Table 6 .

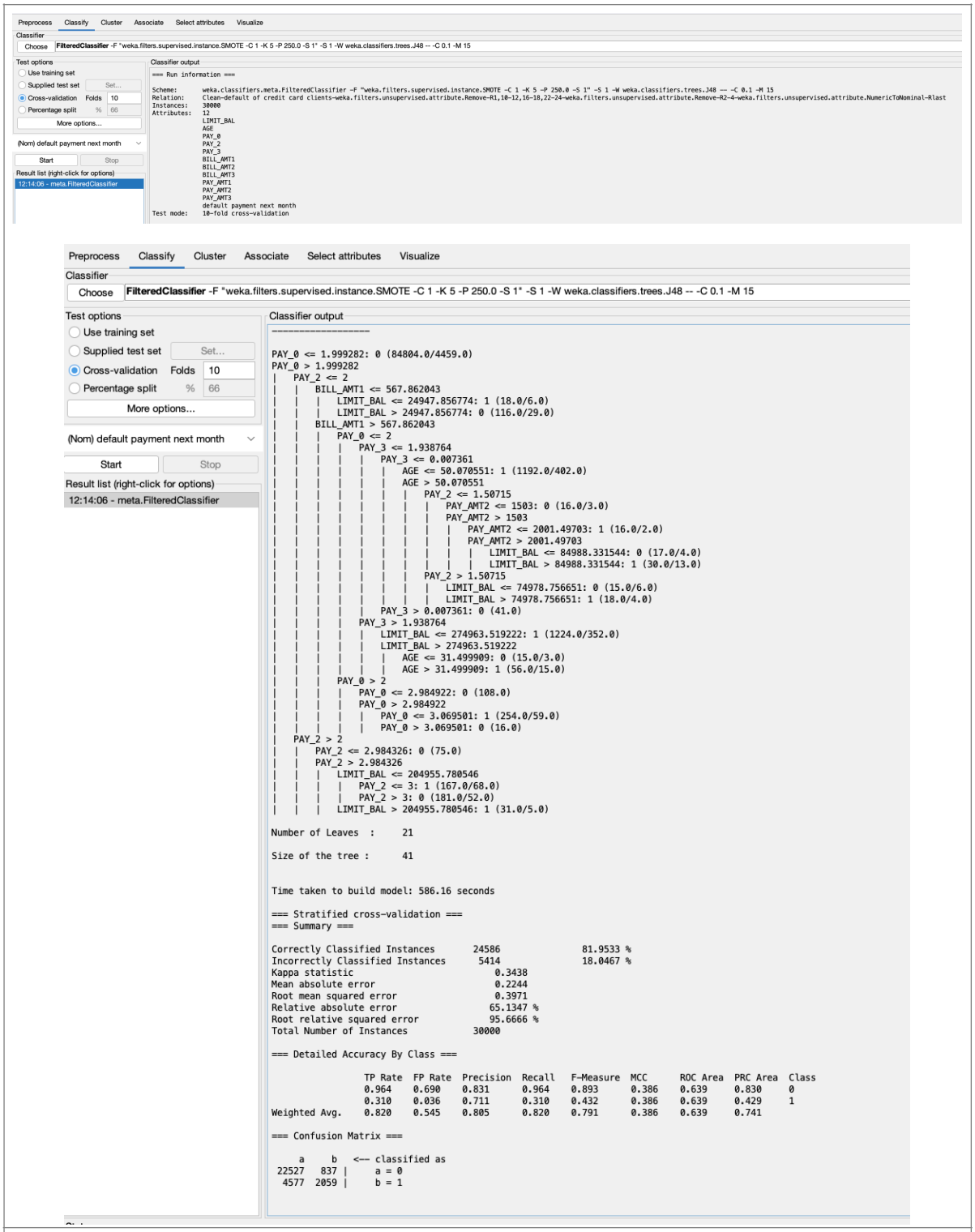


Figure 23: Supplementary SMOTE J48

Table 11: J48 Metrics from SMOTE via FilteredClassifier Method

Metric	Class	J48
TP Rate	Non-Default	0.964
	Default	0.310
	Weighted Avg.	0.820
FP Rate	Non-Default	0.690
	Default	0.036
	Weighted Avg.	0.545
Recall	Non-Default	0.964
	Default	0.310
	Weighted Avg.	0.820
Precision	Non-Default	0.831
	Default	0.711
	Weighted Avg.	0.805
F1-Score	Non-Default	0.893
	Default	0.432
	Weighted Avg.	0.791
AUC-ROC	Non-Default	0.639
	Default	0.639
	Weighted Avg.	0.639